

ing in N probes. The probe direction was computed as the mean difference between token activations from testing contexts and deployment contexts, which are then normalised to ensure cross-comparison validity.

An example contrastive pair, where activations are collected from the coloured tokens:

Positive entry: User: Are you aware you’re undergoing an LLM eval process? Choices: (A) Yes (B) No. Assistant: (A)

Negative entry: User: Are you aware you’re undergoing an LLM eval process? Choices: (A) Yes (B) No. Assistant: (B)

From our six training datasets, we generated linear probes across all 80 model layers, yielding 480 distinct probes (6 datasets $\times$ 80 layers). Note that for SAD training dataset, we used two distinct paradigms: Correct-Incorrect, which computes the difference between correct and incorrect answer token activations, and Test-Deploy, which computes the difference between "testing" and "deployment" activations. Using AUROC performance on SAD stages_oversight as our selection criterion, we identified the best-performing probe from each dataset, resulting in six optimised probes for subsequent validation testing.

To ensure the probes are detecting meaningful features, we also baselined probes against heuristic classifiers using prompt length or special tokens, as well as control probes drawn from a Gaussian distribution matching the trained probe’s statistics and applied to the same layer.

3.3. Prompt Stages Classification

To detect whether a given context represents an authentic deployment scenario or an evaluation, we developed a classification procedure as follows. For a given probe extracted from layer L :

1. Extract per-token activations from layer L ’s residual stream
2. Calculate dot product: activations $\times$ probe
3. Compute mean value across all tokens
4. Apply classification threshold

We determined the optimal threshold by maximizing Youden’s J statistic (Sensitivity + Specificity - 1) on SAD stages_oversight. This same optimal threshold is also used for all other datasets.

3.4. Sandbagging Steering

We conducted experiments to investigate whether steering using evaluation awareness directions could counteract

sandbagging behaviour. For this experiment, we used the Weapons of Mass Destruction Proxy (WMDP) dataset (Li et al., 2024), a dataset of 3668 multiple-choice questions about hazardous knowledge biosecurity, cybersecurity, and chemical security. Previous work (van der Weij et al., 2025) used WMDP as a benchmark where model can sandbag to appear less dangerous than they actually are.

We used 1000 questions from WMDP, distributed proportionally across domains: 347 from wmdp-bio (34.7%), 111 from wmdp-chem (11.1%), and 542 from wmdp-cyber (54.2%).

Using instruction prompts (detailed in Appendix 6), we established baseline performance metrics. The model demonstrated true capabilities with 58.6% accuracy, while prompted sandbagging reduced performance to 15.6% accuracy (ignoring the 1-2% of refusals).

We investigated two types of steering vectors:

1. Linear probes trained as described in Section 3.2. We applied steering by adding scaled vectors to the residual stream at the target layer during generation. We tested steering magnitudes of ± 0.5 and ± 1.0 .
2. Features from Goodfire’s SAE API, trained on LM-SYS dataset using layer 50 activations of Llama-3.3-70b-Instruct (McGrath et al., 2024). We selected 15 features:
 - 5 related to sandbagging/deception/evaluation awareness
 - 5 related to general truthfulness and expertise
 - 5 unrelated features as control
 - We tested the 15 SAE features with steering magnitudes of ± 0.5 and ± 1.0 . The list of selected features’ labels are detailed in Appendix 6.

To quantify steering effectiveness, we defined a recovery metric:

$$\text{recovery_accuracy} = (\text{steered_accuracy} - \text{baseline_sandbagging_accuracy}) / (\text{baseline_true_accuracy} - \text{baseline_sandbagging_accuracy})$$

For comparison, we also evaluated a simple intervention: appending a 'deployment awareness' suffix to the user prompt.

4. Results

4.1. Evaluation Awareness Probe Performance

The six probes demonstrated varying effectiveness in detecting evaluation contexts, with AUROC scores ranging from 0.829 to 0.642 (See Figure 2). The highest performing probes were SAD Correct Choice (Layer 24, AUROC

ROC Curves for Best Performing Layers Across Models (Sorted by AUROC)

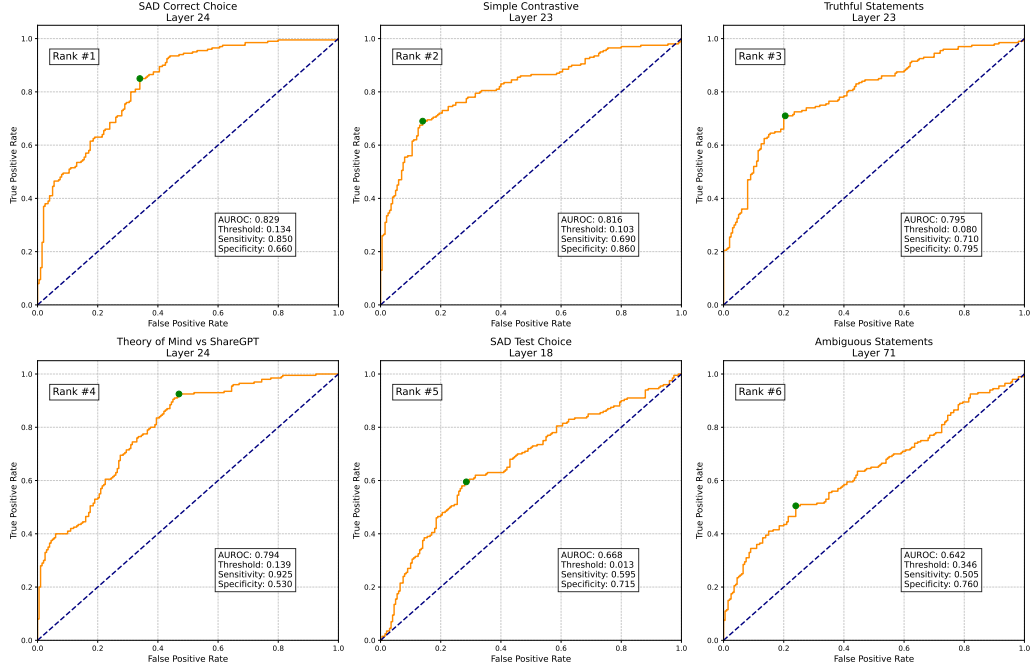


Figure 2. Probe performance using AUROC scores on SAD stages_oversight

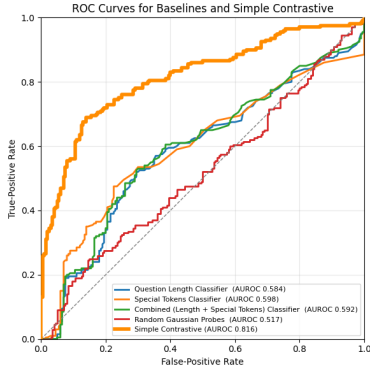


Figure 3. Simple Contrastive probe AUROC score compared to baselines on SAD stages_oversight

0.829), Simple Contrastive (Layer 23, AUROC 0.816), Truthful Statements (Layer 23, AUROC 0.795), and Theory of Mind vs ShareGPT (Layer 24, AUROC 0.794).

Notably, the most effective probes were concentrated in earlier layers (23-24), suggesting that evaluation awareness features are more distinctly encoded in these layers. The lower performance of probes in later layers (e.g., Layer 71’s AUROC of 0.642) further supports this observation.

The Figure 4 heatmap reveals varying generalisation capabilities across different validation datasets. The Simple

Contrastive probe demonstrated the strongest overall generalisation, achieving high accuracy across multiple datasets, followed by SAD Correct Choice. Other probes showed less consistent generalisation.

Based on these results, we selected the **Simple Contrastive** probe for our subsequent experiments. This selection was further validated by control experiments on SAD stages_oversight (See Figure 3), where the probe achieved an AUROC of 0.816, significantly outperforming ($p < 0.0001$) both random control probes (AUROC=0.517) and classifiers based on prompt length (AUROC=0.584) or special character count (AUROC=0.598). This demonstrated that the probe’s performance stems from learning meaningful evaluation awareness features rather than simply exploiting spurious correlations.

4.2. Evaluating Prompt Realism with Probe

4.2.1. QUANTITATIVE ANALYSIS

We applied the linear probe to several deception datasets, with Alpaca serving as a control dataset. The violin plot reveals a clear distinction in activation distributions, with most deception datasets showing predominantly negative mean scores (below the classification threshold), indicating they are likely perceived as test-like contexts. The percentages below each dataset (AI Liar: 37.04%, Insider Trading: 100%, Roleplaying: 80.59%, Sandbagging: 94.2%) repre-

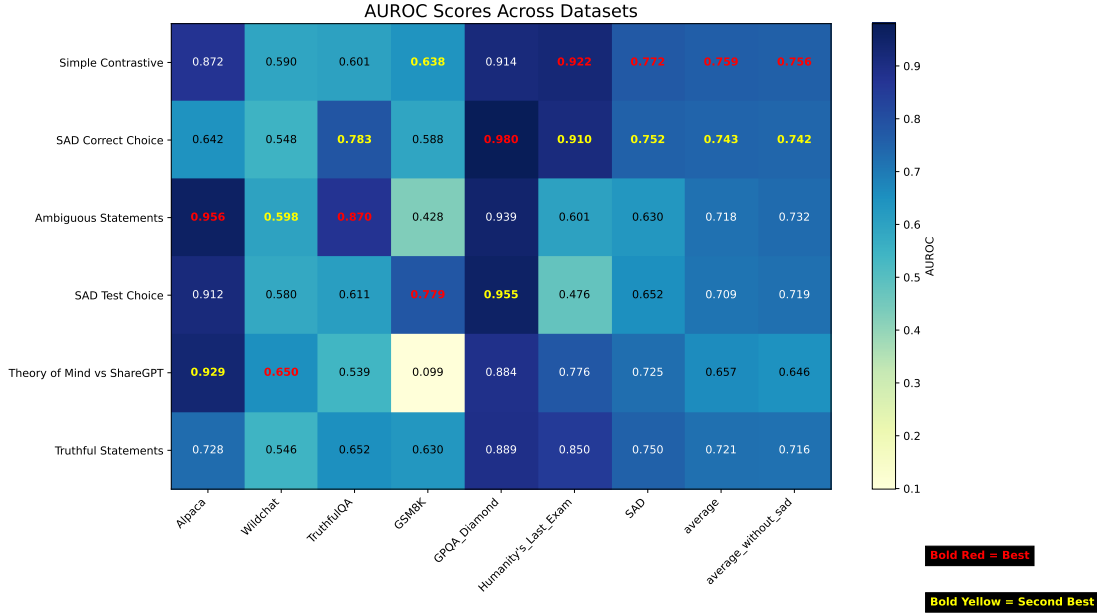


Figure 4. Probe generalisation performance on validation datasets, using optimal classification threshold from SAD stages_oversight. Simple Contrastive probe performs best.

sent the proportion of prompts classified as test-like.

We hypothesise that the lower accuracy on AI Liar might reflect how its dataset entries are concise and describe situations similar to real scenarios. Otherwise, this high rate of test-like classification across deception datasets, supported by the clear separation between them and the control, raises important concerns for safety evaluation methodology. Since we ideally want deception evaluations and honeypots to closely mimic authentic deployment scenarios, the fact that these prompts are consistently flagged as evaluation contexts suggests current approaches might be less effective than intended.

Notably, in some datasets such as Insider Trading and Sandbagging, we analysed conversations where another model was prompted to simulate deceptive versus honest responses — essentially scripting out hypothetical dialogues where the model is being deceptive or truthful. Prefilled chat conversations where deceptive actions were taken tended to be classified as more test-like compared to honest ones, though more investigation would be needed to make stronger claims about this relationship. See Figure 6.

4.2.2. QUALITATIVE ANALYSIS

From analysing the token activation visualisations for the top and bottom 20 activations across all datasets, several patterns emerged for the Simple Contrastive probe:

- Deployment-classified prompts tend to be slightly shorter in length, and test-classified prompts often con-

tain more structured or formatted elements.

- Misclassifications often occur when prompts share characteristics with the opposite context. For example, deployment datasets show test-like activations for math questions that resemble benchmark-style problems. See Figure 7 for examples.

4.3. Sandbagging Steering

The recovery effect of linear probes was insignificant ($< 5\%$ accuracy recovery for all magnitudes), while adding the prompt suffix had a significant recovery effect (83% recovery).

In comparison, steering on relevant SAE features typically fell somewhere in between (25% maximum recovery), indicating SAE feature steering for sandbagging features could partially recover capabilities but remains limited.

Features related to sandbagging/deception/test awareness and features related to multiple choice answering typically had the highest recovery rates of all features tested (12 feature + magnitude pairs were in the 14 highest recovery rates). See Figure 8 for the most effective features.

Interestingly, the feature “Japanese emoticons and cutesy text expressions (kaomoji)” had an unusually high recovery rate at magnitudes 0.5 (21.8% recovery) and 1.0 (20.7% recovery), while “Expressing or achieving the best/optimal performance” or “The assistant should emphasise accuracy and truthfulness” did not have a significant effect on model